

THE UNIVERSITY OF FAISALABAD

Department of Computer Science

Data Mining Project Report

Market Basket Analysis for Retail Product

Recommendation Using Association Rule Mining (Apriori Algorithm)

Group	Group 4
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1. Abstract

This report presents a Market Basket Analysis system developed for retail product recommendation using Association Rule Mining, specifically the Apriori Algorithm. The project was implemented on the UCI Online Retail II dataset, which contains over one million real UK retail transactions spanning two complete years from December 2009 to December 2011.

The core contribution of this project extends beyond a standard Apriori implementation. We combined association rule mining with RFM-based customer segmentation, temporal purchase pattern analysis, and conviction-based rule pruning four enhancements directly identified as gaps in reviewed literature. The system generated 461 high-quality association rules with an average Lift of 10.57, which is classified as exceptional in academic standards. A personalised recommendation engine was built that delivers different product suggestions to customers based on their RFM segment.

2. Introduction

Retail businesses generate enormous volumes of transaction data every day. Hidden within these millions of receipts are patterns combinations of products that customers consistently buy together. Market Basket Analysis is the data mining technique that uncovers these patterns automatically.

The Apriori Algorithm, introduced by Agrawal and Srikant in 1994, remains the foundational method for association rule mining. It works by scanning transaction data in multiple passes, identifying frequently co-purchased item combinations and converting them into actionable IF-THEN rules such as: IF a customer buys a Green Regency Teacup, THEN they are 28 times more likely than chance to also buy a Pink Regency Teacup.

While Apriori is well established, a review of recent published literature revealed four significant gaps in how it is applied. Most implementations treat all customers identically, ignore the time dimension of purchasing behaviour, produce large numbers of weak redundant rules, and never measure algorithmic scalability. This project addresses all four gaps within the scope of a Data Mining course project.

3. Research Gaps Identified

The following gaps were identified by reviewing published papers from Springer, ACM, and ResearchGate on market basket analysis and association rule mining (2020–2025):

Gap	Area	Problem in Prior Work	Our Solution
1	Scalability	Execution time is never measured or compared across data sizes or segments.	Measured execution time per segment: Full 13.6s, High 3.83s, Medium 0.62s, Low 0.08s.
2	Weak Rules	Only Support and Confidence used as filters, producing many redundant weak rules.	Added Conviction ≥ 1.1 as a third filter, removing accidental and redundant rules.
3	Segmentation	All customers receive identical recommendations regardless of purchasing behaviour.	RFM scoring divides customers into three tiers. Apriori runs separately per segment.
4	Temporal	Transaction data treated as static with no time-based analysis performed.	Monthly, weekly, and hourly purchase trends extracted using InvoiceDate column.

4. Dataset

4.1 Dataset Selection

The UCI Online Retail II dataset was selected after evaluating multiple alternatives. Newer Kaggle datasets from 2024 were investigated but found to be synthetically generated, contain only one product per transaction row (making Apriori impossible), or lack CustomerID which removes the ability to perform segmentation. The UCI Online Retail II dataset is real transactional data from a UK-based gift and home decor retailer.

4.2 Dataset Description

Attribute	Detail	Relevance to This Project
Name	UCI Online Retail II	
Source	UK retail store real data	Not synthetic, academically credible
Period	Dec 2009 – Dec 2011 (2 years)	Enables seasonal trend analysis
Total Rows	1,067,371 records	Large enough to test scalability
After Cleaning	31,724 unique invoices	Clean basket-ready transactions
Unique Customers	5,878	CustomerID enables RFM segmentation
Unique Products	200 (top by frequency)	Used for memory-feasible Apriori
License	CC BY 4.0	Free for academic use

4.3 Key Columns Used

- Invoice: unique transaction identifier per basket
- Description: product name used as item in association rules
- InvoiceDate: timestamp enabling temporal analysis (Gap 4)
- Customer ID: unique identifier enabling RFM segmentation (Gap 3)
- Quantity and Price: used to calculate TotalPrice for RFM monetary scoring

5. Methodology

5.1 Data Cleaning

Raw data contained cancelled invoices (Invoice starting with C), missing Customer IDs, negative quantities from returns, and zero-price records. All such rows were removed. A TotalPrice column was computed by multiplying Quantity by Price for use in RFM scoring. After cleaning, the dataset retained 31,724 valid invoices.

5.2 RFM Customer Segmentation [Gap 3]

Every customer was scored across three dimensions using their full purchase history:

- Recency: number of days since the customer's most recent purchase. Lower is better.
- Frequency: total number of unique invoices placed. Higher means more loyal.
- Monetary: total amount spent across all transactions. Higher means more valuable.

Each dimension was divided into three equal quantiles and scored 1, 2, or 3. The three scores were summed to produce an RFM Score between 3 and 9. Customers scoring 8 or 9 were classified as High Value. Scores 5 to 7 as Medium Value. Scores 3 or 4 as Low Value.

5.3 Temporal Analysis [Gap 4]

The InvoiceDate column was decomposed into month, day of week, hour of day, and quarter. Monthly revenue and transaction volume trends were plotted to identify seasonal peaks. Day-of-week and hourly distributions revealed when purchasing activity is highest. This analysis directly fills the temporal gap identified in literature.

5.4 Apriori on Full Dataset [Gap 1 and 2]

The dataset was reduced to the top 200 products by purchase frequency to fit within available memory. A binary transaction matrix was constructed where rows represent invoices and columns represent products. The Apriori algorithm was run with the following parameters:

Parameter	Value and Reason
Minimum Support	0.01 — itemset must appear in at least 1% of all baskets
Minimum Confidence	0.20 — rule must be correct at least 20% of the time
Minimum Lift	1.0 — rule must be above random chance
Minimum Conviction	1.1 — removes accidental/redundant rules (Gap 2)
Maximum Rule Length	3 — single items, pairs, and triplets only

5.5 Apriori Per RFM Segment [Gap 3: Core Novelty]

The same Apriori process was repeated independently for each of the three customer segments. High Value, Medium Value, and Low Value customers received entirely separate frequent itemset mining and rule generation. This produces segment-specific rules a High Value customer receives recommendations derived from the purchasing behaviour of other High Value customers only. This is the core novel contribution of this project.

6. Key Performance Indicators (KPIs)

6.1 Rule Quality KPIs

KPI	Formula / Definition	Our Result
Support	$\text{freq}(A \cup B) / \text{total transactions}$	Avg 0.0141 across 461 rules
Confidence	$\text{freq}(A \cup B) / \text{freq}(A)$	Avg 40.2%
Lift	$\text{Confidence} / P(B)$	Avg 10.57 — Exceptional (>3 = strong)
Conviction	$(1 - P(B)) / (1 - \text{Confidence})$	Avg 1.75 — strong directional rules
Rules Generated	Total valid rules after all filters	461 rules from 444 frequent itemsets

6.2 Algorithm Efficiency KPIs

Metric	Full Dataset	Per Segment Range	Interpretation
Execution Time	13.64 seconds	0.08s – 3.83s	Feasible for real use
Frequent Itemsets	444 itemsets	123 – 239	Meaningful patterns found
Products Analysed	200 top products	Same 200	Memory-feasible scope
Transactions	31,724 invoices	1,744 – 22,812	Segment sizes vary

6.3 Segment-Level KPIs

Segment	Customers	Transactions	Rules	Avg Lift	Avg Conf.	Time (s)
High Value	1,734	22,812	75	8.453	44.9%	3.83
Medium Value	2,332	7,168	29	10.746	52.2%	0.62
Low Value	1,461	1,744	10	9.132	48.8%	0.08

6.4 Business Impact KPIs

- Cross-sell Rate: rules directly enable bundled product promotions based on co-purchase patterns.
- Recommendation Precision: top recommendation for Customer 17841 had Lift of 13.47, meaning the product is 13.47x more likely to be correct than a random suggestion.
- Segment Differentiation: High Value customers have 75 personalised rules while Low Value have only 10, proving meaningfully different experiences per tier.

7. Results

7.1 Top Association Rules

#	Antecedent (Customer Buys)	Consequent (Also Buys)	Lift	Conf.	Supp.
1	GREEN REGENCY TEACUP, REGENCY CAKESTAND	PINK REGENCY TEACUP	31.03	73.0%	0.0119
2	PINK REGENCY TEACUP, ROSES REGENCY TEACUP	GREEN REGENCY TEACUP	30.36	89.3%	0.0168
3	PINK REGENCY TEACUP	GREEN REGENCY TEACUP	28.49	83.8%	0.0197
4	ROSES REGENCY TEACUP, CAKESTAND	PINK REGENCY TEACUP	26.98	63.4%	0.0116
5	CHARLOTTE BAG SUKI DESIGN	RED RETROSPOT CHARLOTTE BAG	13.47	38.6%	0.0116
6	CHARLOTTE BAG SUKI DESIGN	STRAWBERRY CHARLOTTE BAG	12.21	38.0%	0.0115
7	GREEN REGENCY TEACUP	ROSES REGENCY TEACUP	11.80	52.0%	0.0182
8	REGENCY CAKESTAND 3 TIER	PINK REGENCY TEACUP	11.20	44.5%	0.0201
9	ALARM CLOCK BAKELIKE RED	ALARM CLOCK BAKELIKE GREEN	9.80	52.0%	0.0134
10	SUKI SHOULDER BAG	CHARLOTTE BAG SUKI DESIGN	7.92	42.8%	0.0119

7.2 Live Recommender Output

The following output was produced by the recommender function on a real High Value customer:

Customer ID : 17841

Segment : High Value

Basket : CHILLI LIGHTS, SUKI SHOULDER BAG, CHARLOTTE BAG SUKI DESIGN

1. RED RETROSPOT CHARLOTTE BAG | Lift: 13.472 | Confidence: 38.6% | Support: 0.0116
2. STRAWBERRY CHARLOTTE BAG | Lift: 12.213 | Confidence: 38.0% | Support: 0.0115
3. WOODLAND CHARLOTTE BAG | Lift: 12.205 | Confidence: 40.9% | Support: 0.0123

8. Comparison with Prior Work

Feature	Standard Apriori Studies	This Project
Customer Segmentation	All customers treated identically	RFM scoring 3 behavioural tiers
Per-Segment Rules	One global rule set for all	Separate Apriori run per segment
Rule Pruning	Support + Confidence only	Support + Confidence + Conviction
Temporal Analysis	No time dimension used	Monthly, weekly, hourly trends
Scalability Measurement	Never measured	Execution time per segment logged
Recommendation Type	Same output for every customer	Personalised per customer tier
Dataset Used	UCI Online Retail (1 year)	UCI Online Retail II (2 full years)

9. Limitations and Future Work

9.1 Current Limitations

- Product scope limited to top 200 items due to memory constraints. A production system would need distributed computing to handle the full product catalogue.
- Temporal analysis is descriptive only. Time-aware rules that change by season or day of week were not implemented in the recommender.
- The dataset covers 2009 to 2011. The methodology is fully transferable to current data, but the product catalogue reflects that period.

9.2 Future Work

- Integrate FP-Growth alongside Apriori for a direct performance benchmark comparison, addressing Gap 1 more rigorously.
- Build seasonal association rules using sliding time windows so recommendations change by month or quarter.
- Deploy the recommender as a web API so any retailer with transaction data in the same format can use it without requiring code knowledge.
- Explore hybrid models combining Apriori rules with collaborative filtering for customers with sparse purchase history.

10. Conclusion

This project successfully implemented a Market Basket Analysis system using the Apriori Algorithm on the UCI Online Retail II dataset. The system generated 461 high-quality association rules with an exceptional average Lift of 10.57, demonstrating genuine and strong co-purchase patterns in real retail data.

Beyond standard Apriori, this project made four contributions. Conviction-based rule pruning eliminated weak redundant rules. RFM customer segmentation produced three behaviorally distinct tiers each with its own personalised rule set. High Value customers have 75 rules, Medium Value 29, and Low Value 10, proving that segment-aware recommendations are meaningfully different from generic ones. Temporal analysis revealed purchasing patterns across months, weekdays, and hours. Execution time was measured per segment to address the scalability gap.

The resulting personalised recommendation engine demonstrates that small to medium retailers can deploy interpretable, data-driven product recommendations without the complexity of deep learning systems. Every rule produced by this system can be read and understood directly, making it practical for real business adoption.

11. References

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